

MODEL CARD - Simple PINN

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1 Overview

The purpose of this project was to implement a basic physics-informed neural network (PINN) to solve a toy model problem. In doing so, the hope was to create a guide for the implementation of PINNs using PyTorch. Using a PINN with only a few hundred trainable parameters, the network was able to learn the dynamics of cooling governed by Newton’s law of cooling. Compared to the baseline model, which did not incorporate a physics-informed loss, the PINN generalised beyond the supervised observations in the temporal domain by learning solutions that satisfied the underlying governing differential equation, rather than simply fitting the available data.

2 Technical Details

Compared to typical feed-forward neural networks, PINNs incorporate an additional physics-informed loss term derived from the governing equations of the physical process. The difference is not primarily architectural, but rather its optimisation. Given our toy problem, that of a cooling object governed by a single differential equation (Newton’s law of cooling), we only have a single PDE residual. The simplicity of our problem allows us to avoid enforcing the initial condition $T(t = 0) = T_0$ within the loss function $\mathcal{L}$,

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \cancel{\lambda_{\text{con}}\mathcal{L}_{\text{con}}}$$

For more complicated dynamics, boundary and initial conditions must be into into $\mathcal{L}_{\text{con}}$.

As mentioned previously, there is only a single PDE (now ODE) residual δ_T used in our physics-informed loss term $\mathcal{L}_{\text{PDE}}$. Its form is,

$$\delta_T = \frac{dT}{dt} + \alpha(T - T_s)$$

In some PINNs, the physics-informed loss is calculated using only collocation points. This is the case in our example. To be explicit, collocation points are points which are sampled across the entire domain but lack a corresponding label. Collocation points are unsupervised and used to enforce the PDE (ODE in our example) dynamics over the entire domain. It should be noted that in our problem, we make a distinction between collocation and data points because the labelled data has been sampled from a restricted domain. This was done to showcase that PINNs can generalise to unseen data by encoding physics through unsupervised learning. The physics-informed loss is constructed from the squared residual averaged over all randomly drawn collocation points during each batch iteration. Our ODE loss term is written as,

$$\mathcal{L}_{\text{PDE}} = \frac{1}{N_c} \sum_{c=1}^{N_c} \left(\frac{d\hat{T}}{dt_c} + \alpha[\hat{T}(t_c) - T_s] \right)^2$$

where N_c is the number of collocation points per batch and $\hat{T}(t_c)$ is the model’s temperature prediction at the collocation point t_c . N_c is kept fixed across all batch iterations of the optimisation process and is an additional hyperparameter that must be tuned accordingly. Note that during each batch, the collocation points are randomly drawn from the entire temporal domain. With this in mind, the full physics-informed loss function $\mathcal{L}$ can be written as,

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N [T(t_i) - \hat{T}(t_i)]^2 + \frac{\lambda}{N_c} \sum_{c=1}^{N_c} \left(\frac{d\hat{T}}{dt_c} + \alpha[\hat{T}(t_c) - T_s] \right)^2$$

where λ is a hyperparameter that controls how strongly the network considers the ODE residual. It is important to distinguish the difference between $T(t_i)$, which is the model prediction on a supervised data point t_i that has the associated label $T(t_i)$, versus the model prediction $\hat{T}(t_c)$ on the collocation point t_c .

3 Technical Details

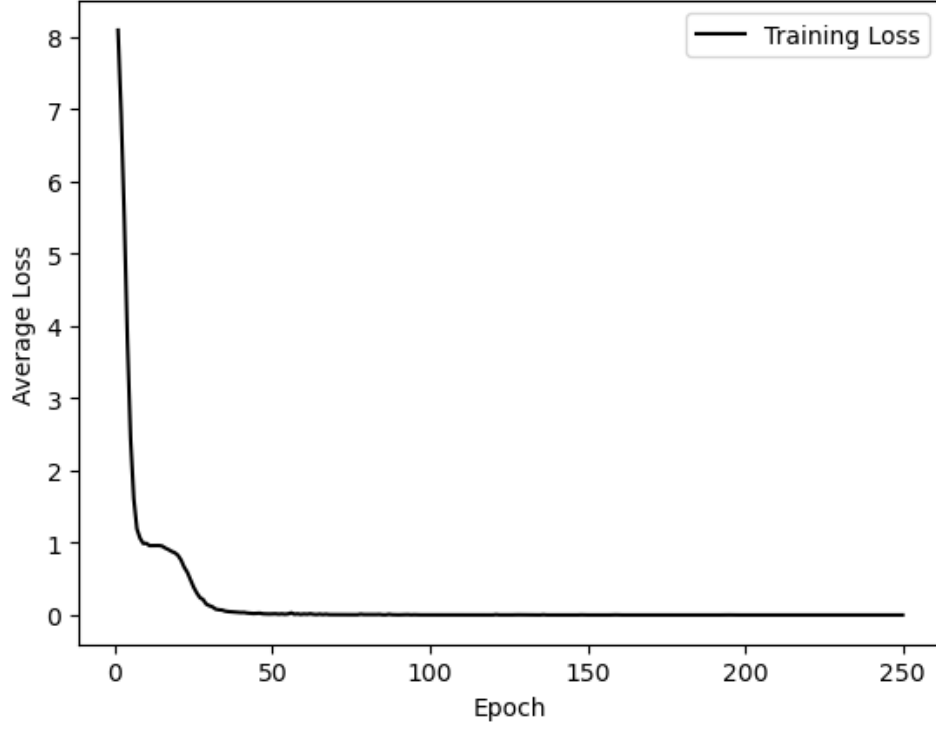


Figure 1: Training the PINN.

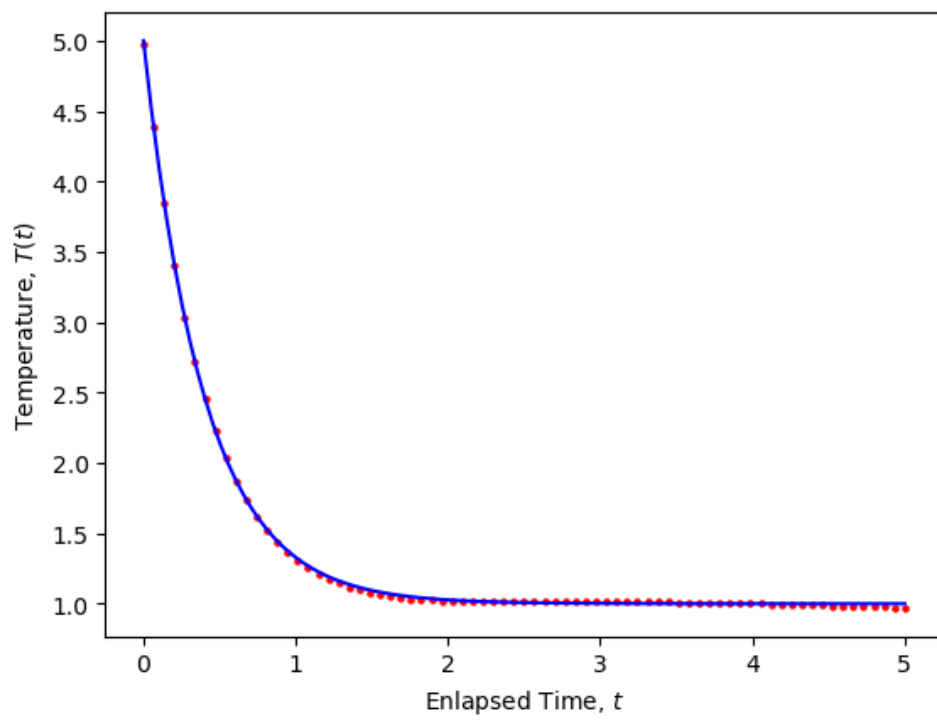


Figure 2: The PINN is able to generalise.